Guidelines - Assignment 1

Files provided:

A1_Assignment.pdf	Instructions and guidelines
A1_Notebook.ipynb	Notebook you must complete
A1_Template.docx	Three-page reporting template
South_German_Credit.csv	South German Credit dataset

Group work:

Work in **teams of two students**. You can find a teammate on the Toledo course discussion board.

When working on the assignment:

- Complete the supplied **A1_Notebook.ipynb in Python,** using Google Colab or another notebook editor. Keep all code and relevant outputs in this notebook.
- Fill in **A1_Template.docx** and add both names and r-numbers. Keep its three-page layout and font size. Keep answers concise and figures and tables legible.
- AI tools**, such as ChatGPT,** may assist with coding. You remain responsible for checking and understanding the code, calculations and conclusions.

You will have to submit both the completed notebook and a report.

To submit your assignment:

1. Prepare your files. Run all notebook cells in order, check the outputs and save. Export the completed report as a PDF.

2. Name your files and email. Replace **r1** and **r2** with your actual r-numbers, smaller number first.

Item	Use this format	Example
Email subject	DSFB_A1_r1_r2	DSFB_A1_r0123456_r0765432
Report (PDF)	A1_r1_r2.pdf	A1_r0123456_r0765432.pdf
Notebook	A1_r1_r2.ipynb	A1_r0123456_r0765432.ipynb

3. Send your email. Please send one email per team to **andreas.goethals@kuleuven.be** with both files attached. Include both names and r-numbers in the message.

Support:

Use the Toledo course discussion board (preferred), or email your questions with subject **DSFB_A1_QUESTION_r1_r2**.

Submissions not respecting the provided guidelines (e.g., template or format) will receive a zero grade.

Deadline: 10 November 2026 at 23:59
Belgian local time (Europe/Brussels)

Situation and task:

As consultants at **Leuven Analytics Experts & Partners (LAE)**, help a regional bank explore credit risk, build a logistic regression model and explain it through a scorecard.

Dataset: South German Credit

South_German_Credit.csv contains **1,000 credit records from 1973-1975**, with **20 predictors and one target (21 columns)**. There are 700 good and 300 bad credits, with no missing values.

Use the **South German Credit** codes below. **DM** = German marks.

Variable in the CSV	Meaning, units or category codes
QUANTITATIVE PREDICTORS (3) - measured values	
duration	Credit duration in months.
amount	Credit amount after an undocumented monotonic transformation; not an interpretable amount in DM.
age	Age in years.
CATEGORICAL PREDICTORS (17) - numbers are category codes	
status Checking account	**1:** none; **2:** balance < 0 DM; **3:** 0 ≤ balance < 200 DM; **4:** balance ≥ 200 DM or salary paid in for ≥ 1 year.
credit_history Repayment history	**0:** past payment delays; **1:** critical account / other credits elsewhere; **2:** no credits taken / all credits repaid duly; **3:** existing credits paid duly so far; **4:** all credits at this bank repaid duly.
purpose Credit purpose	**0:** other; **1:** new car; **2:** used car; **3:** furniture/equipment; **4:** radio/TV; **5:** domestic appliances; **6:** repairs; **7:** education (not observed); **8:** vacation; **9:** retraining; **10:** business.
savings Savings balance (DM)	**1:** unknown / no savings account; **2:** < 100 DM; **3:** 100 to < 500 DM; **4:** 500 to < 1,000 DM; **5:** ≥ 1,000 DM.
employment_duration Years with current employer	**1:** unemployed; **2:** < 1 year; **3:** 1 to < 4 years; **4:** 4 to < 7 years; **5:** ≥ 7 years.
installment_rate % of disposable income	**1:** ≥ 35%; **2:** 25 to < 35%; **3:** 20 to < 25%; **4:** < 20%.
personal_status_sex Personal status and sex	**1:** male, divorced/separated; **2:** female, non-single OR male, single; **3:** male, married/widowed; **4:** female, single. Sex cannot be separated from this variable.
other_debtors	**1:** none; **2:** co-applicant; **3:** guarantor.
present_residence	Years at current residence: **1:** < 1; **2:** 1 to < 4; **3:** 4 to < 7; **4:** ≥ 7.
property Most valuable property	**1:** unknown / none; **2:** car or other; **3:** building society savings agreement / life insurance; **4:** real estate.
other_installment_plans	Other instalment plans: **1:** bank; **2:** stores; **3:** none.
housing	**1:** free accommodation; **2:** rented; **3:** owned.
number_credits	Credits at this bank, including the current one: **1:** 1; **2:** 2-3; **3:** 4-5; **4:** ≥ 6.
job Employment type	**1:** unemployed/unskilled, non-resident; **2:** unskilled, resident; **3:** skilled employee/official; **4:** manager/self-employed/highly qualified employee.
people_liable	Financial dependants: **1:** 3 or more; **2:** 0 to 2.
telephone	Landline registered in the customer's name: **1:** no; **2:** yes.
foreign_worker	Foreign-worker status: **1:** yes; **2:** no.
credit_risk	Target: **0:** bad credit; **1:** good credit.

Source: South German Credit (2020), UCI, codetable.txt and read_SouthGermanCredit.R; corrected documentation by Ulrike Grömping (2019). CC BY 4.0. CSV values and row order unchanged; English variable names used.

Question 1 - Explore the data

The bank wants to understand how applicant and loan characteristics relate to bad credit. Explore the data and explain one useful pattern.

1. Load and check the data

Load **South_German_Credit.csv** into your Python notebook. Check the dimensions, variable types, missing values, duplicate rows and target counts. Create a modelling target with **1 = bad and 0 = good**; keep the original credit_risk column. Record your checks and any action taken in the notebook.

2. Keep a test sample separate

Before selecting predictors or exploring their relationship with the target, make a reproducible, stratified **70% training / 30% test split**. Record the random seed. Use this split throughout the assignment: training data for exploration and model development, test data only for final evaluation.

3. Choose two variables and make one figure

Choose **two variables** and justify your choices. Define groups using the combination of both variables, grouping quantitative values or combining categories where appropriate. Make **one clearly labelled figure** showing each group's **bad-credit rate and number of observations**. Use descriptive category labels and explain your grouping choices.

4. Interpret the evidence

Identify the group with the highest bad-credit rate and the group containing the most bad credits. Discuss the reliability of the pattern, considering group sizes and your grouping choices. Give one useful takeaway for the bank.

In the report - Part 1

Briefly explain your variable and grouping choices, insert your figure, and give **at most four bullet points** interpreting it. Keep the data checks, split details and supporting calculations in the notebook.

Question 2 - Build and evaluate the model

Select predictors and use WOE to build and assess one logistic regression model for bad credit.

1. Select and transform your predictors

Select predictors from page 2, including at least one quantitative variable. **Decide how many predictors to include and justify your final selection.** You may reuse predictors from Question 1. Define suitable levels for each predictor by binning quantitative values or combining categories where appropriate. Transform every selected predictor using **weights of evidence (WOE)**. **Describe and justify your predictor and level choices in the report.**

2. Fit and interpret one model

Fit a logistic regression model using **only your selected WOE-transformed predictors**.

Among the selected predictors, which has the highest training **information value (IV)**? Explain what its WOE values tell you about which levels are more or less risky.

3. Evaluate the model

Plot the **training and test ROC curves together in one figure,** with their AUC values in the legend. Explain how well the model distinguishes good from bad observations and whether the difference suggests overfitting. Give one conclusion for the bank and one limitation.

In the report - Part 2

Justify your final predictor set and its level definitions, insert the ROC figure, and answer the interpretation questions in **at most four bullet points**. Keep the selection evidence, WOE values, IV values and fitted coefficients and intercept in the notebook.

Question 3 - Construct the scorecard

Convert your model from Question 2 into a scorecard: a table of points that can be added to obtain a credit score.

1. Construct and check the scorecard

Create the full scorecard with columns **predictor, level and points,** using the approach discussed in the lectures. Use these scale specifications:

- A score of **600** corresponds to **good:bad odds of 50:1**.
- Each **20-point increase doubles the good:bad odds** (PDO = 20).

Higher scores therefore indicate lower predicted risk. For every test **observation,** compare the total from your scorecard with the score implied by its model probability. Verify both scale specifications. Keep your derivation, checks and explanation of any differences in the notebook.

2. Visualize the scores

Create **one figure with two boxplots** of the test scores: one for observations with a good outcome and one for observations with a bad outcome. Highlight these two cases:

- The **highest-scoring bad observation**.
- The **lowest-scoring good observation**.

In the report - Part 3

Include the **full scorecard,** followed by the boxplot figure. Keep the derivation and verification in the notebook.